

CS 230 : Discrete Computational Structures

Fall Semester, 2023

HOMEWORK ASSIGNMENT #2

Due Date: Tuesday, September 12

Suggested Reading: Rosen Sections 1.4 - 1.6; LLM Chapter 3

For the problems below, explain your answers and show your reasoning. Even if you get a problem wrong, you are more likely to get partial credit if you show your work.

1. [8 Pts] For the following problems, let $R(x)$, $C(x)$ and $D(x, y)$ be the statements “ x has read the notes”, “ x is in your class” and “ x has discussed problems with y ”, respectively. Assume that $D(x, x)$ evaluates to false.

Let the domain be all people at ISU. Translate from English to logic.

- (a) Someone in your class has read the notes but has not discussed problems with anyone in your class.
 - (b) There are two students in your class who have, between them, discussed problems with everyone in the class.
2. [15 Pts] Define predicates and prove the following using the appropriate rules of inference:
 - (a) [5 Pts] Mary, a student in class, has visited Paris. Everyone who visits Paris goes to the Arc de Triomphe. Therefore, someone in class has visited the Arc de Triomphe.
Universe: all ISU students
 - (b) [5 Pts] There are college towns in the midwest. All college towns are fun places to live. There is a town in the midwest that is a fun to live in.
Universe: all towns
 - (c) [5 Pts] All bears are good swimmers. If you can catch fish, you will not go hungry. If you can't catch fish, you are not a good swimmer. Therefore, no bears go hungry.
Universe: all animals
 3. [5 Pts] Prove, by giving a counterexample, that the propositions $\exists x(P(x) \wedge Q(x))$ and $\exists xP(x) \wedge \exists xQ(x)$ are not logically equivalent. *Hint:* Let your domain be the set of integers. Define $P(x)$ to be ‘ x is odd’ and $Q(x)$ to be ‘ x is even’.
 4. [12 Pts] For this problem the universe is $\mathcal{N} = \{0, 1, 2, 3, \dots\}$. We will construct predicates using logical operations and quantifiers. Also, assume that you are given the predicates

$A(n, m, k)$, which means $n + m = k$ and $M(n, m, k)$, which means $n * m = k$.

We can define new predicates from these such as $Zero(n) = A(n, n, n)$ that will return TRUE if $n = 0$ and FALSE otherwise.

Since we now have the predicate $Zero(n)$, we can use this to build other predicates. For example, $Greater(m, n)$, which means $m > n$, can be defined as $\exists k \neg Zero(k) \wedge A(n, k, m)$.

Note that we defined $Zero(n)$ only in terms of a previously defined predicate; we did not introduce a new predicate (based on a mathematical relation) in the definition, as in “ $Zero(n) = (n = 0)$,” and we invoked predicates using variables for each argument, not numbers as in “ $Zero(n) = A(n, 0, 0)$.” Likewise, we defined $Greater(m, n)$ only in terms of previously defined predicates, a quantifier, and logical relations; we did not introduce a separate predicate based on a mathematical relation, as in “ $Greater(m, n) = (m > n)$.”

For each of the parts below, define new predicates for each that uses quantifiers, logical operations, and predicates given above or ones that you have already constructed (in a previous part).

Note that you cannot use mathematical relations in your definitions, and the arguments for the predicates must be variables, not specific numbers (e.g. an expression like “ $A(n, m, m)$ ” is ok but an expression like “ $M(n, 5, 10)$ ” is not).

For partial credit, give a short explanation of how you would construct the predicate.

- (a) [4 pts] $Equal(m, n)$ that is true if and only if $m = n$
- (b) [4 pts] $One(n)$ that is true if and only if $n = 1$
- (c) [4 pts] $Two(n)$ that is true if and only if $n = 2$
- (d) [4 pts Extra Credit] $Prime(p)$ that is true if and only if p is prime

For more practice, work on the problems from Rosen Sections 1.4 - 1.6 and LLM Chapter 3.